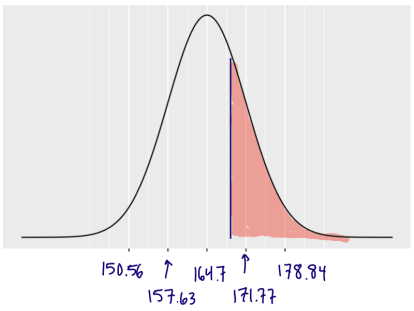
A

1. (a) appliances sold
   (b) time until scrapped
   (c) $H_0: \mu = 15$ vs. $H_a: \mu < 15$.

2. (a) III (b) II (c) IV (d) I

4. (a) 1 - pnorm(168, 164.7, 7.07)

   (b)



   (c) qnorm(0.82, 164.7, 7.07)

   (d) $\sigma = 7.07$ and so $SE_{\bar{x}} = \frac{7.07}{\sqrt{25}} = 1.414$

   pnorm(168, 164.7, 1.414)

5. (a) μ is the average weight of these oranges considering the entire population.

   (b) qt(0.965, 15)

   (c) $149.2 \pm (1.95094)\frac{9.2}{\sqrt{16}} = 149.2 \pm (1.95094)(2.3) = 149.2 \pm 4.487$

   or $(144.71, 153.69)$

   (d) Option (i)

6. (a) $\hat{p} = \frac{33}{72} \doteq 0.4583$

   (b) It is valid, since $np = (72)(0.5) = 36$ and $n(1-p) = (72)(0.5) = 36$. Both are at least 5.

   (c) $z = \frac{\hat{p} - p}{SE_{\hat{p}}} = \frac{0.4583 - 0.5}{\sqrt{(0.5)^2/72}} = \frac{-0.0417}{0.05893} \doteq -0.7076$

   (d) pnorm(-0.7076)

   (e) Since the p-value is larger than $\alpha = 0.05$, we fail to reject the null hypothesis. Our data is consistent with the possibility that half of city residents support having no restrictions.